Pearson Correlation Test

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Introduction

The Pearson correlation test assesses whether two continuous variables are linearly related. It returns a correlation coefficient r and a p-value for the null that the population correlation $\rho = 0$, along with a confidence interval computed via Fisher's z transformation.

Prerequisites

Pearson correlation coefficient, sampling distributions.

Theory

Given iid pairs (X_i, Y_i) from a bivariate distribution with marginal SDs σ_X, σ_Y, the sample correlation is

$$r = \frac{\sum(X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum(X_i - \bar{X})^2 \sum(Y_i - \bar{Y})^2}}.$$

Under $H_0 : \rho = 0$, the statistic

$$t = r\sqrt{\frac{n-2}{1-r^2}} \sim t_{n-2}.$$

For confidence intervals, Fisher's z-transformation is

$$z = \frac{1}{2}\log\frac{1+r}{1-r}, \quad \mathrm{SE}(z) = \frac{1}{\sqrt{n-3}}.$$

Construct a CI in the z scale, then back-transform with $r = \tanh(z)$.

Assumptions

- Continuous X and Y, approximately bivariate normal.
- Linear relationship; no extreme outliers.
- Independent pairs.

R Implementation

```r
library(broom)
set.seed(2026)

n <- 80
x <- rnorm(n)
y <- 0.4 * x + rnorm(n, sd = sqrt(1 - 0.4^2))  # true rho = 0.4
```

```r
cor.test(x, y, method = "pearson") |> tidy()

# Manual check
r <- cor(x, y)
t_stat <- r * sqrt((n - 2) / (1 - r^2))
p_val  <- 2 * (1 - pt(abs(t_stat), df = n - 2))
c(r = r, t = t_stat, p = p_val)

# Fisher's z CI
z    <- atanh(r)
se_z <- 1 / sqrt(n - 3)
ci_z <- z + c(-1, 1) * 1.96 * se_z
tanh(ci_z)
```

Output & Results

```
# A tibble: 1 x 8
  estimate statistic p.value parameter conf.low conf.high method    alternative
     <dbl>     <dbl>   <dbl>     <dbl>    <dbl>     <dbl> <chr>     <chr>
1    0.43      4.19   <0.001        78    0.234     0.598  Pearson   two.sided
```

Sample $r = 0.43$ (95 % CI 0.23 to 0.60), $t(78) = 4.19$, $p < 0.001$. Strong evidence of a positive correlation.

Interpretation

"There was a significant positive correlation between x and y ($r = 0.43$, 95 % CI 0.23 to 0.60; $t(78) = 4.19$, $p < 0.001$). The two variables shared 18.5 % of their variance linearly."

Practical Tips

- Scatter plot first; r can be misleading if the relationship is non-linear.
- Robust alternatives for heavy tails or outliers: Spearman, Kendall.
- For small n, the CI via Fisher's z is noticeably asymmetric in the r scale.
- Always report both r and its CI; the p-value alone is often uninformative.
- Testing multiple correlations from the same data requires multiple-testing correction.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors
- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds

- Pearson and Spearman correlation
- Non-parametric tests